

Birmingham City University



Tutorial 2: Audio in MATLAB

Samuel Smith

Task 1: Getting to Grips with MATLAB

Have a look at the *matlab_intro.m* file in the materials on Moodle. This is a quick overview of MATLAB syntax.

Complete the following things in MATLAB:

1. Write a for loop which will display the first 10 powers of 2.
2. Write a function to add two numbers together and return the result.
3. Write a function to calculate the root mean square amplitude of an array of values.

$$\text{RMS}(x) = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} (x[n])^2}$$

Remember that MATLAB uses 1-based indexing when you are iterating through the array.

Task 2: Loading and Manipulating Audio

Have a look at the *import_wav.m* file in the materials on Moodle. This is a short MATLAB script which loads a .wav file and performs some simple analysis/manipulation.

Using this code as a starting point, find some of your own audio files online and write MATLAB code to do the following things:

1. Plot the waveform of a signal.
2. Apply gain to a signal by multiplying it by a constant.
3. Apply DC offset to a signal by adding a constant.

4. Normalise a signal to have a peak amplitude of 1.
5. Normalise a signal to have an RMS amplitude of 0.5.
6. Apply a fade in/fade out effect to some audio. (look up the *linspace()* and *logspace()* functions in MATLAB, along with the element-wise multiplication operator `.*`).
7. Reverse a signal.
8. Swap the left and right channels of a stereo file.
9. Sum a stereo file to mono.
10. Load multiple audio files and mix them together by summing them.
11. Do a remix: cut out portions of an audio file and reorder them.

To experience the true wonder of your creations:

- Investigate the *sound()* function in MATLAB to play them out the speakers (if online this will only work when using certain browsers).
- Investigate the *audiowrite()* function in MATLAB to save them to an audio file.

Task 3: Synthesising Audio

Have a look at the *create_signal.m* and *generate_sinusoid.m* files in the materials on Moodle. Here we use a function to synthesise a sinusoid according to a set of parameters.

Using this code as a starting point, complete the following:

1. Investigate the effects of aliasing: what combinations of sampling frequency and signal frequency produce valid outputs?
2. Generate sinusoidal signals of different frequencies and sum them together.
3. Concatenate several signals to make a short tune.
4. Generate some binaural beats: play sine waves of slightly differing frequencies out the left and right channels of a stereo signal.